The association between EEG alpha band power and postoperative delirium in patients undergoing hip fracture surgery

A mechanistic cohort study within a randomised controlled trial of intraoperative infliximab

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INTRODUCTION

Recent evidence suggests one potential pattern in resting state electroencephalography (rsEEG) prior to induction of anaesthesia that may indicate vulnerability to delirium is power in the alpha band oscillations.[1] Early work indicates this feature is seen in major elective surgery.[2] However, this cohort focused on relatively healthier older adults undergoing non-urgent, major surgery.[3] In contrast, the majority of the burden of postoperative delirium is amongst frail, multimorbid patients undergoing emergency surgery such as those with a hip fracture.[4]

Our aim was to clarify mechanistic links between EEG markers of cerebral vulnerability and postoperative delirium in frail elderly surgical patients. We hypothesised that reduction in power in the alpha band in rsEEG is predictive of cerebral sensitivity to the effects of anaesthesia which give rise to increased vulnerability to the development of postoperative delirium after hip fracture surgery.

METHODOLOGY

We conducted a mechanistic cohort study embedded within the DECI trial, a phase III double blinded, two-arm, parallel, multicentre, randomized controlled trial of intraoperative infliximab infusion (5mg.kg-1) versus equal-volume saline infusion to reduce postoperative delirium. Participants aged over 60 years undergoing hip fracture surgery were eligible. Participants with a contraindication to infliximab therapy or refusal for EEG recordings were ineligible. Written informed consent was obtained from each participant, or legal proxy (next of kin or professional consultee) in patients who lacked capacity to consent. Research ethics committee approval for the study was obtained (20/SC/0452) and study registration performed before recruitment commenced (ISRCTN28447330).

The primary trial outcome was delirium, assessed using the Memorial Delirium Assessment Scale over five postoperative days by research staff blinded to the DECI allocation and EEG data. The primary EEG outcome of interest was relative alpha-band power in rsEEG.

EEG recordings were obtained in the operating theatre before induction of anaesthesia until end of surgery using a standardised 4-channel scalp montage (F7, Fp1, Fp2, and F8) using Sedline (Masimo Inc., Irvine, CA). Impedance was kept <10kΩ . Analysis was performed in MATLAB with the EEGLAB plug-in. Data were sequentially notch-, high- and low pass filtered (48-52Hz, 1Hz, and 40Hz respectively, 4th-order zero phase Butterworth). Recordings were visually inspected for bad channels and cleaned using sliding window principal component analysis with artificial subspace reconstruction and re-referenced to a common average. Multitaper power spectral density estimates were calculated using discrete prolate spheroidal (Slepian) sequences (windows 2s, half-bandwidth 1.5 Hz, 5 tapers). Mean band powers for each participant were calculated prior to induction in the eyes-closed condition. Multivariable linear regression models were fitted for the association between delirium severity and relative alpha band power, adjusting for age, sex and pre-existing cognitive impairment.

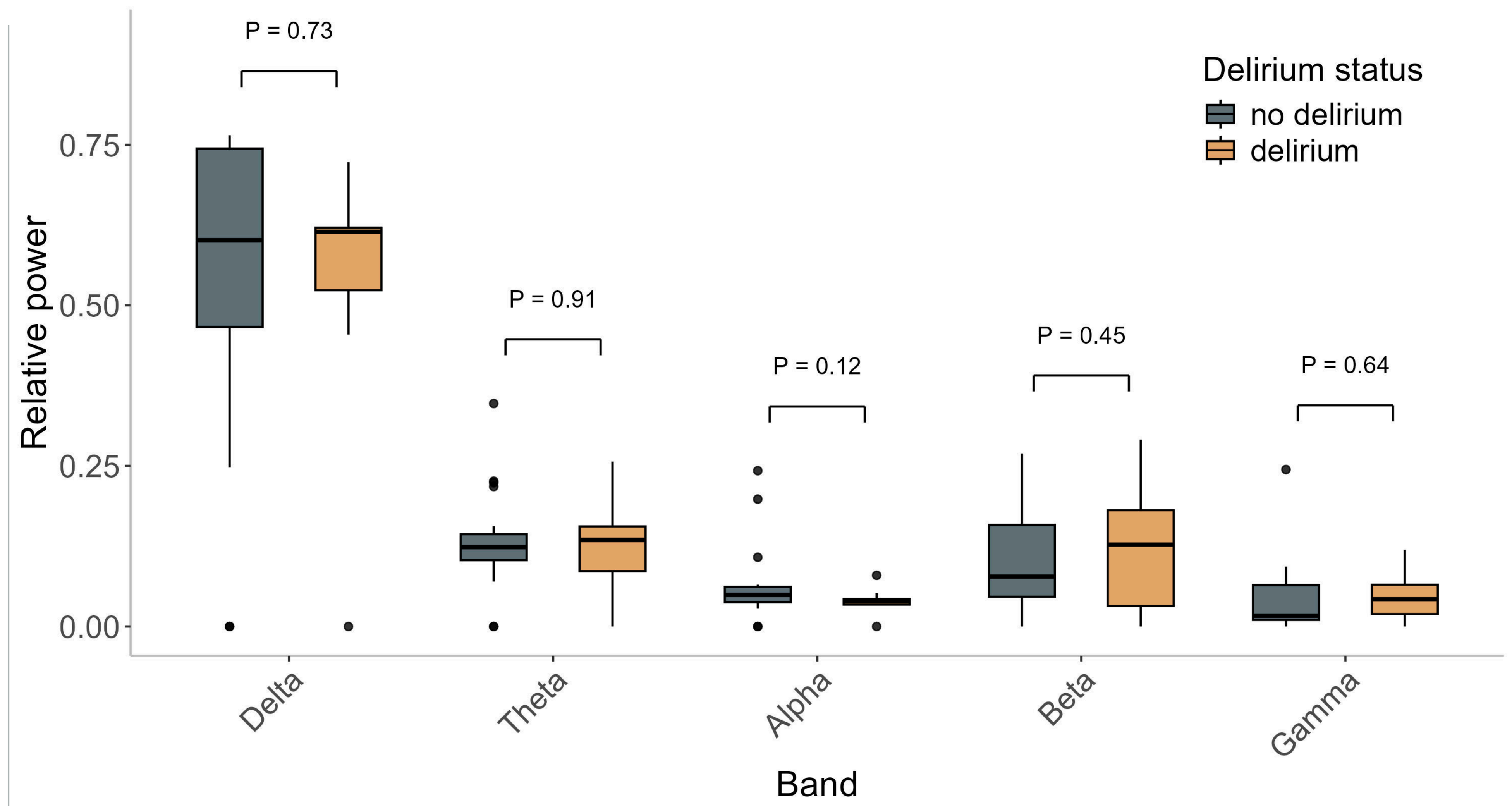

Figure 1. Mean relative bandpower by postoperative delirium status (measured on Memorial Delirium Assessment Scale). Delta band [1 to 4Hz], Theta band [4 to 8Hz], Alpha band [8 to 13Hz], Beta band [13 to 30Hz], Gamma band [30 to 40Hz].

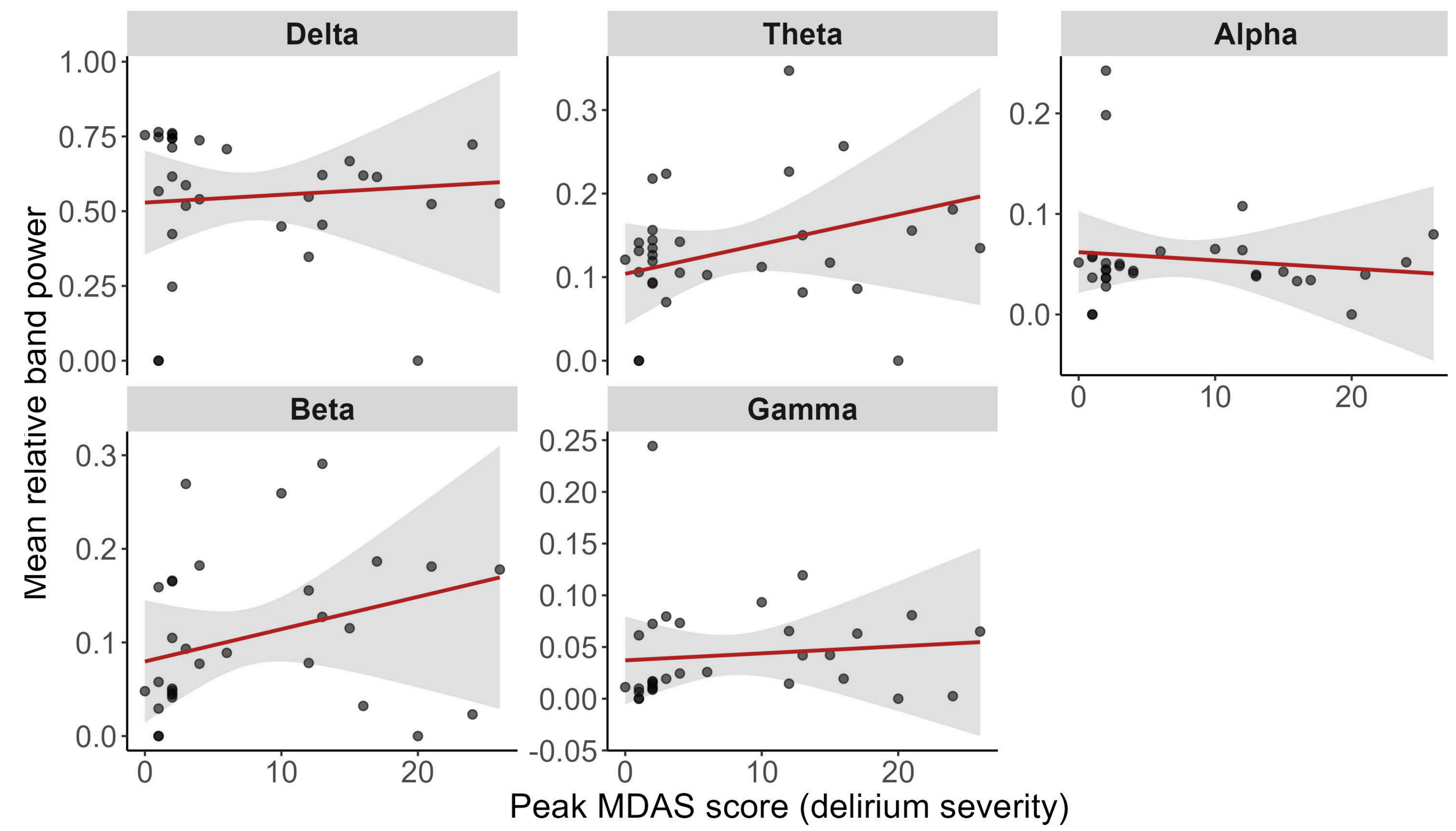

Figure 2. Adjusted association between delirium severity and relative band power

RESULTS

Between July 2024 and September 2025 we recruited 34 participants for the mechanistic EEG cohort study within the DECI trial. The incidence of postoperative delirium was 26% (n=9/34). During data cleaning 3 participants' recordings were removed for excessive noise. There was no significant difference in crude relative alpha power between patients with and without postoperative delirium (mean difference -0.022, 95%CI -0.051 to 0.005, p=0.12, unpaired Student's t-test). Mean relative band power in rsEEG for all bands are shown in Figure 1. Across all bands, there was no significant association in between relative band power on rsEEG and postoperative delirium incidence.

Mean relative alpha band power was not significantly associated with postoperative delirium severity after adjustment for covariates (p=0.721) (Figure 2). Pre-existing cognitive impairment was associated with delirium severity (p<0.001).

CONCLUSIONS

In patients undergoing hip fracture surgery, we found that relative alpha power in the eyes closed condition prior to induction of anaesthesia was not associated with postoperative delirium. These findings suggest that cerebral vulnerability to postoperative delirium is not manifest as reduced frontal alpha power prior to emergency orthopaedic trauma surgery in elderly frail patients.

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